Vicente Bastura

Mechanical Engineer
vicentebastura1310@gmail.com | (860) 481-1231

Education

Worcester Polytechnic Institute – BS/MS in Mechanical Engineering | GPA – 3.82 May 2027
Extracurriculars – Combat Robotics, Running Club, FSAE

Projects

Autonomous Off-Road Vehicle Chassis – Major Qualifying Project May 2026 - Present
- %ï Contributed to iterative spaceframe chassis design by considering structural, packaging,
- %ï Conducted static FEA simulations using Ansys to analyze the chassis for torsional rigidity
- %ï Collaborating with other teams to create an entire off-road vehicle platform, with a full ro

1 Pound Plastic Combat Robot October 2025 - Present
- %ï Designed a parametric weapon, iterated mechanical components, and collaboratively brain
- %ï Manufactured the complete robot using 3D printing and wired the electronic system for th
- %ï Identified weaknesses in workflow, material, and design considerations for version 2 deve
- %ï Redesigned the failed weapon to be asymmetric and smaller for version 2.

MIDI Music Box August 2025 - October 2025
- %ï Collaborated on the mechanical design, assembly of custom 3D-printed actuators, and an music box.
- %ï Wired and integrated 20 servos with servo driver boards to independently actuate individu
- %ï Developed Arduino code to parse MIDI input and map incoming notes to the corresponding
- %ï Implemented individual servo position offsets in software to compensate for variation in t actuation.

Understanding Perceptions of Cuenca Cantón Wildfires – Interactive Qualifying Project October 2025 - March 2026
- %ï Collaborated with a 4-person team and the University of Cuenca to investigate community Cuenca, Ecuador.
- %ï Interviewed rural and municipal stakeholders alongside University of Cuenca students, ad populations.
- %ï Performed statistical analysis and authored the results section, applying chi-squared test respondents' survey data.

Skills

CAD and Design: SolidWorks (CSWA & Simulation), Ansys, Fusion 360, Onshape, 3D Part & Assembly Design, 2D Drawings
Manufacturing and Prototyping: 3D Printing (Operation & Maintenance), Rapid Prototyping, CAM, CNC Mill, CNC Lathe, Calipers, Micrometer, Manufacturing Process, Osha-10 Certified, Tolerancing
Programming and Electronics: MATLAB, Arduino (C++), Raspberry Pi, Sensors, Actuators, Soldering
Other: Computer Skills, Problem Solving, Attention to Detail, Hands-On, Creative, Teamwork, Communication, AI

Experience

Landscaping, Multiple Companies – Pomfret, CT May 2024 – August 2026
- %ï Managed a private client's property independently, scheduling work, and determining mai assigned projects including pathway and retaining-wall construction. (2024-2026)
- %ï Completed daily tasks independently at a golf course, operating push mowers, weed trimm (2024)
- %ï Maintained residential properties for a landscaping company, traveling between job sites